Active Learning in Undergraduate STEM Courses: A Literature Review of Achievement, Equity and Student Perception

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Abstract

This literature review synthesises seven studies of active learning in undergraduate science, technology, engineering and mathematics (STEM) courses published between 1998 and 2020. Three themes organise the evidence: effects on achievement and failure rates, effects on achievement gaps for underrepresented students, and the mismatch between how much students learn and how much they feel they learn. Across meta-analyses and course-level studies, active learning raises exam and concept-inventory scores by roughly half a standard deviation, lowers failure rates and narrows achievement gaps, but students often rate it below a polished lecture. The review closes by identifying gaps in dosage, long-term retention and replication beyond physics and biology.

1 Introduction

Lecturing has been the default mode of undergraduate STEM instruction for more than a century, yet a growing body of research argues that students learn more when they spend class time solving problems, answering conceptual questions and discussing their reasoning with peers (Prince, 2004; Freeman et al., 2014). This literature review examines what that research has established, for whom the benefits hold, and why students do not always recognise them. The scope is limited to undergraduate STEM courses in which an active-learning format was compared with lecture-based instruction on a measured learning outcome.

Sources were identified by searching ERIC, Web of Science and Google Scholar for the terms *active learning*, *interactive engagement* and *peer instruction* combined with *undergraduate* and *STEM*, *physics* or *biology*, and by following citations from the two meta-analyses that the search returned. Studies were included if they (1) were published in a peer-reviewed venue between 1998 and 2020, (2) involved undergraduate STEM courses, and (3) reported an empirical comparison with lecture-based instruction or a quantitative synthesis of such comparisons. Seven studies met these criteria; Table 1 summarises them.

2 Thematic Review

2.1 Achievement and failure rates

The earliest large-scale evidence comes from Hake (1998), who collected pre- and post-test data from 62 introductory physics courses with 6,542 students. Using the normalised gain $\langle g \rangle$ on the Force Concept Inventory and the earlier Mechanics Diagnostic test, he found that the 14 traditional courses averaged $\langle g \rangle = 0.23 \pm 0.04$ while the 48 courses that made substantial use of interactive-engagement methods averaged 0.48 ± 0.14, almost two standard deviations higher. Prince (2004) reviewed the same body of work for engineering faculty and concluded that there was "broad but uneven support"for the core elements of active, collaborative, cooperative and problem-based learning. The most comprehensive estimate is the meta-analysis by Freeman et al. (2014), which pooled 225 studies across STEM disciplines. Active learning raised examination and concept-inventory scores by 0.47 standard deviations, and the mean failure rate fell from 33.8 % under traditional lecturing to 21.8 % under active learning, so students in lecture courses were about 1.5 times more likely to fail. The effect was present in every discipline examined and was largest in classes of fifty or fewer students.

2.2 Equity: who benefits most

A second line of research asks whether active learning narrows the achievement gaps that persist between well-represented and underrepresented students. Eddy and Hogan (2014) compared three terms of a low-structure introductory biology course with three terms of a moderate-structure version taught by the same instructor, which added guided reading, graded preparatory homework and in-class activities. The moderate-structure design raised exam performance for all students by about 3 %, but Black students gained roughly twice as much, halving the gap with white students, and first-generation students closed the gap with continuing-generation peers. Theobald et al. (2020) generalised this result in a meta-analysis of student-level data from 15 studies of exam scores (9,238 examinations) and 26 studies of passing rates (44,606 students): active learning narrowed achievement gaps by 33 % on exam scores and by 45 % on passing rates. The narrowing was concentrated in courses with high-intensity active learning, defined as at least two thirds of class time spent on active work, which led the authors to propose that gaps close only when deliberate practice is combined with inclusive teaching.

2.3 Student perception and implementation

The third theme concerns how students experience active learning. Crouch and Mazur (2001) reported ten years of Peer Instruction in introductory physics courses for non-majors at Harvard, documenting gains in both conceptual reasoning and quantitative problem solving and describing refinements such as replacing in-class reading quizzes with pre-class written responses. Deslauriers et al. (2019) then isolated the question of perception with a randomised crossover experiment in which 149 students in a calculus-based physics course were taught one topic actively and another by lecture, with identical content and slides. Students in the active condition scored almost half a standard deviation higher on a test of learning (standardised coefficient 0.46) but rated their own learning more than half a standard deviation lower (-0.56). The authors attribute the gap to the greater cognitive effort that active work demands, which novices read as a sign that they are learning less, and they recommend that instructors explain the method early in the term.

Table 1: Studies included in the review.

Study	Design	Sample	Main finding
Hake (1998)	Multi-course pre/post survey	62 physics courses, 6,542 students	Interactive engagement roughly doubled the normalised gain (0.48 vs. 0.23).
Crouch and Mazur (2001)	Ten-year course case study	Harvard introductory physics for non-majors	Peer Instruction improved conceptual reasoning and problem solving.
Prince (2004)	Narrative review	Engineering education literature	Broad but uneven support for active, collaborative, cooperative and problem-based learning.
Eddy and Hogan (2014)	Quasi-experiment, six terms	One introductory biology course, about 393 students per term	Moderate structure raised all scores and halved the Black–white gap.
Freeman et al. (2014)	Meta-analysis	225 studies across STEM	Scores up 0.47 SD; failure rate 21.8 % vs. 33.8 %.
Deslauriers et al. (2019)	Randomised crossover	149 physics students	Active sections learned more but felt they learned less.
Theobald et al. (2020)	Meta-analysis of student-level data	15 studies (exams), 26 studies (pass rates)	Gaps narrowed by 33 % on exams and 45 % on passing rates.

3 Methodological Critique

The strength of this literature lies in its convergence: a six-thousand-student survey, two meta-analyses, a six-term course comparison and a randomised experiment point in the same direction. Its weaknesses are also consistent. First, most primary studies are quasi-experimental, comparing different cohorts or sections rather than randomly assigned students, so differences in instructor enthusiasm or student self-selection cannot be ruled out; only Deslauriers et al. (2019) randomised at the student level, and that study covered two class meetings rather than a semester. Second, "active learning"is defined loosely. Freeman et al. (2014) accepted any format in which students did more than listen, whereas Theobald et al. (2020) found that only high-intensity formats narrowed gaps, which suggests that pooled effect sizes hide large differences in dosage. Third, outcome measures vary from concept inventories to course grades; Freeman et al. (2014) themselves note that effects are larger on concept inventories than on course examinations, so the headline effect size depends on what is counted as learning. Finally, the evidence is concentrated in introductory physics and biology at research universities in the United States, and several of the key studies share authors and institutions.

4 Gaps and Future Directions

Three gaps follow from this critique. The first is dosage: no study in the set varies the proportion of class time given to active work systematically, so the threshold at which benefits appear is unknown. The second is retention: outcomes are measured at the end of a class or a term, and there is little evidence on whether the gains persist into later courses or into professional practice. The third is generalisation beyond the dominant settings, including community colleges, mathematics

and engineering courses, and institutions outside North America. Future research should combine student-level randomisation with semester-long interventions, report the intensity of active learning explicitly, and follow students across subsequent courses.

5 Conclusion

This literature review set out to establish what is known about active learning in undergraduate STEM courses, for whom it works and how students perceive it. The evidence reviewed supports three claims. Active learning improves achievement and lowers failure rates relative to lecturing, with an effect of about half a standard deviation that holds across disciplines (Hake, 1998; Freeman et al., 2014). Its benefits are largest for students from underrepresented groups, but only when the active component is substantial and the classroom is deliberately inclusive (Eddy and Hogan, 2014; Theobald et al., 2020). And students systematically underestimate how much they learn from it, which means that course evaluations are an unreliable guide to effectiveness (Deslauriers et al., 2019). The main limitations of the literature are its reliance on quasi-experimental designs, loose definitions of the intervention and short follow-up periods. Research that measures dosage, tracks retention and extends the evidence to new settings would turn a strong general finding into practical guidance about how much active learning is enough.

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